

SHUMIKHA, Russian Federation

WMO: 286550

Lat: 55.233N

Lon: 63.317E

Elev: 582

StdP: 14.39

Time Zone: 5.00 (E05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.1	-16.9	-26.9	1.3	-21.0	-21.8	1.7	-16.8	20.3	21.9	18.4	16.4	3.5	320	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	88.0	69.1	84.6	67.6	81.3	66.2	72.5	83.6	70.2	80.3	68.1	77.9	7.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.6	107.2	79.5	66.5	99.6	76.8	64.3	92.2	73.9	36.5	84.1	34.5	80.4	32.7	77.7	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.3	14.4	-28.2	93.2	6.1	3.2	-32.6	95.6	-36.2	97.5	-39.7	99.3	-44.1	101.6		
(5)			-28.4	74.0	5.8	4.0	-32.6	76.9	-36.0	79.3	-39.2	81.5	-43.5	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	3.9	8.4	21.9	41.3	54.9	64.2	66.3	63.5	52.3	38.9	22.3	10.2
	DBStd	24.37	11.46	11.86	8.92	9.94	9.21	8.28	6.76	8.05	8.19	9.07	12.54	12.86
	HDD50	6318	1429	1166	870	297	57	6	0	4	66	359	831	1233
	HDD65	10344	1894	1586	1335	713	333	114	66	127	388	809	1281	1698
	CDD50	1753	0	0	0	35	208	431	506	422	135	16	0	0
	CDD65	302	0	0	0	0	19	89	106	80	8	0	0	0
	CDH74	3140	0	0	0	15	296	938	999	803	89	0	0	0
	CDH80	971	0	0	0	2	70	304	303	273	19	0	0	0
	WSAvg	6.6	6.4	6.7	7.1	7.7	7.1	6.4	5.7	5.6	6.0	6.6	6.9	6.8
	PrecAvg	16.1	0.7	0.6	0.7	0.7	1.8	1.7	2.8	2.3	1.3	1.3	1.1	1.0
(15) Precipitation	PrecMax	23.0	1.4	1.1	1.6	1.6	4.6	3.4	8.7	5.9	3.3	3.3	2.4	1.9
	PrecMin	9.3	0.2	0.1	0.0	0.0	0.2	0.5	0.4	0.2	0.0	0.1	0.2	0.4
	PrecStd	3.8	0.4	0.3	0.5	0.4	1.3	0.7	1.9	1.6	0.8	0.7	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	36.6	49.7	75.6	86.1	91.0	92.5	91.7	83.0	67.0	46.8	36.4
		MCWB	31.2	33.4	43.5	57.5	65.1	69.8	69.9	70.2	62.2	53.8	42.6	32.6
	2%	DB	29.3	33.6	40.3	68.1	81.6	87.3	86.9	86.9	75.9	62.2	42.7	33.7
		MCWB	27.5	31.0	36.0	52.8	62.6	69.0	70.6	68.2	59.8	52.7	39.9	31.3
	5%	DB	25.7	29.5	37.1	62.7	77.0	83.6	83.2	82.7	70.9	57.1	39.6	30.8
		MCWB	24.2	27.3	33.6	50.6	60.3	67.3	68.9	67.2	59.4	49.7	36.7	29.1
	10%	DB	21.9	25.4	34.7	57.5	72.2	79.6	79.5	78.2	65.7	52.1	36.6	27.6
		MCWB	20.5	23.4	31.8	47.8	57.9	65.4	67.0	65.7	56.9	46.6	34.5	26.1
	0.4%	WB	31.7	34.1	43.6	57.4	68.7	74.0	75.6	74.7	65.4	57.0	43.1	33.6
		MCDB	33.0	35.7	49.7	72.9	82.3	86.6	85.8	86.2	73.1	64.1	45.7	36.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.7	30.9	36.4	54.0	64.8	71.4	72.7	71.4	62.2	53.1	40.1	31.6
		MCDB	29.3	33.4	39.5	65.2	78.2	83.4	83.3	82.6	72.1	60.6	42.5	33.3
	5%	WB	24.2	27.4	34.2	51.7	61.8	68.9	70.6	68.5	59.6	49.9	37.2	29.2
		MCDB	25.7	29.5	36.6	61.0	74.1	80.3	79.9	78.8	69.9	56.6	39.2	30.9
	10%	WB	20.5	23.5	32.1	48.6	59.2	66.7	68.3	66.1	57.3	47.1	34.7	26.1
		MCDB	21.8	25.4	34.4	57.3	70.3	77.6	76.6	76.1	65.1	51.9	36.3	27.6
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	16.0	15.3	16.6	20.4	19.4	18.2	18.5	16.5	13.2	10.8	12.6
		MCDBR	14.5	15.4	15.0	24.5	27.3	25.3	24.0	25.7	23.8	21.2	11.3	12.8
		MCWBR	13.5	13.6	11.6	13.4	13.1	11.5	10.6	11.0	12.4	13.0	8.9	11.7
	5% WB	MCDBR	14.3	15.6	13.5	23.0	24.9	23.7	21.0	22.7	22.1	20.0	10.8	13.1
		MCWBR	13.5	14.1	10.9	13.5	14.0	12.6	11.3	12.2	12.7	12.9	9.2	12.2
(40) Clear-Sky Solar Irradiance	taub		0.284	0.314	0.368	0.409	0.408	0.405	0.425	0.443	0.384	0.355	0.312	0.283
	taud		2.063	2.038	1.968	2.121	2.222	2.303	2.248	2.163	2.292	2.241	2.154	2.037
	Ebn at Noon		193	232	245	256	265	267	258	243	243	217	181	160
	Edh at Noon		21	31	43	43	41	38	40	41	31	26	19	17
(44) All-Sky Solar Radiation	RadAvg		283	628	1120	1422	1719	1860	1790	1404	951	487	250	179
	RadStd		21	26	70	143	163	177	139	140	79	73	31	23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page